COSC 39: Theory of Computation

Prishita Dharampal

Problem 1. Neighborhoods in Graphs.
Let G be an directed graph, possibly with self-loops (an edge whose head and tail is the same vertex). Define V and E to be the vertex set and edge set of G, respectively. The neighborhood function $N : V \to 2^V$ takes a vertex v and maps it to the subset of vertices that are out-neighbors of v in G. We also define the neighborhood of any subset of vertices S as the union of each individual neighborhood for each vertex in S. In notation,

$$N(S) := \bigcup_{v \in S} N(v).$$

Fix a starting vertex s in V. The k-th neighborhood of s is defined to be $N(N(\cdots N(s)\cdots))$, where $N(\cdot)$ is applied k times.

(a) Describe an algorithm to decide if the following is true: for every vertex t in V, there is an integer k such that the k-th neighborhood of s contains t.

Solution.

(a) Let G be a graph and s be the starting vertex. Then to determine if there exists an integer k such that the k-th neighborhood of s contains t is the same as asking if there exists a directed path of length k (for some $k \in \mathbb{Z}$) from s to t (for all t).

To do this we perform DFS, track visited vertices V_v, and return true iff $|V_v| = |V|$. Note that by definition $s \in N^k(s)$ for some k if there is a path from s to s. Then, we can define a pseudo-function as follows:

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Connected(G, s):
visited = {}
stack = [s]
while stack is not empty:
    v = pop(stack)
    for each out-neighbor u of v:
        if u ∉ visited:
            visited = visited ∪ {u}
            push(stack, u)
return 𝟙[|visited| = |V|]
```

Problem 2. Bit reversal.
Consider the following pairs of functions on strings over alphabet Σ:

$$\text{evens}(w) := \begin{cases} \epsilon & \text{if } w = \epsilon, \\ \text{odds}(x) & \text{if } w = a \cdot x \text{ for some } a \in \Sigma \text{ and } x \in \Sigma^* \end{cases}$$

$$\text{odds}(w) := \begin{cases} \epsilon & \text{if } w = \epsilon, \\ a \cdot \text{evens}(x) & \text{if } w = a \cdot x \text{ for some } a \in \Sigma \text{ and } x \in \Sigma^* \end{cases}$$

For example,

$$\text{evens(penpineappleapplepen)} = \text{epnapeplpn},$$

$$\text{odds(penpineappleapplepen)} = \text{pnieplapee}.$$

We define the bit-reversal function as

$$\text{bitrev}(w) := \begin{cases} w & \text{if } |w| \leq 1, \\ \text{bitrev}(\text{odds}(w)) \cdot \text{bitrev}(\text{evens}(w)) & \text{otherwise.} \end{cases}$$

Prove the following statements:

(a) $|\text{odds}(w)| < |w|$ and $|\text{evens}(w)| < |w|$ for any string w of length at least 2.

(b) $|\text{bitrev}(w)| = |w|$ for any string w.

Solution.

(a) Let w be any string, and let $|w|$ denote is length. We will prove a stronger statement to solve this question. We claim that

$$|\text{odds}(w)| = \left\lceil \frac{|w|}{2} \right\rceil, \quad |\text{evens}(w)| = \left\lfloor \frac{|w|}{2} \right\rfloor.$$

We will prove this using induction.

Base Case:

- If $|w| = 0$ then the statement vacuously holds.
- If $|w| = 1$ then $|\text{odds}(w)| = 1$ and $|\text{evens}(w)| = 0$. Hence, the statements holds.

Induction Hypothesis: For any string w of length $|w| < n$,

$$|\text{odds}(w)| = \left\lceil \frac{|w|}{2} \right\rceil, \quad |\text{evens}(w)| = \left\lfloor \frac{|w|}{2} \right\rfloor$$

Inductive Case: Let $w = abx$, where $a, b \in \Sigma$ and $x \in \Sigma^{n-2}$. That is $|w| = n$. Then,

$$|\text{odds}(w)| = |\text{odds}(abx)| = |a\,\text{evens}(bx)| = |a\,\text{odds}(x)| = 1 + |\text{odds}(x)|$$

Note that since $|x| < |w|$ by IH $|\text{odds}(x)| = \left\lceil \frac{|x|}{2} \right\rceil$, and since $|w| = |x| + 2$,

$$\left\lceil \frac{|w|}{2} \right\rceil = \left\lceil \frac{|x|}{2} + 2 \right\rceil = \left\lceil \frac{|x|}{2} \right\rceil + 1 = |\text{odds}(x)| + 1 = |\text{odds}(w)|$$

Similarly,

$$|\text{evens}(w)| = |\text{odds}(bx)| = |b\,\text{evens}(x)| = 1 + |\text{evens}(x)|$$

Note that since $|x| < |w|$ by IH $|\text{evens}(x)| = \left\lfloor \frac{|x|}{2} \right\rfloor$, and since $|w| = |x| + 2$,

$$\left\lfloor \frac{|w|}{2} \right\rfloor = \left\lfloor \frac{|x|}{2} + 2 \right\rfloor = \left\lfloor \frac{|x|}{2} \right\rfloor + 1 = |\text{evens}(x)| + 1 = |\text{evens}(w)|$$

Hence, the induction hypothesis holds, and $|\text{odds}(w)| < |w|$ and $|\text{evens}(w)| < |w|$ for any string w of length at least 2.

(b) We will also prove this using induction. Let w be any string.

Induction Hypothesis: For any string w of length $|w| < n$, $|\text{bitrev}(w)| = |w|$.

Base Case: If $|w| \leq 1$ then by definition,

$$|\text{bitrev}(w)| = |w|$$

Inductive Case: If $|w| = n$, then

$$|\text{bitrev}(w)| = |\text{bitrev}(\text{odds}(w)) \cdot \text{bitrev}(\text{evens}(w))|$$

From part (a), we know that for any string w of length $|w| = n$,

$$|\text{odds}(w)| = \left\lceil \frac{|w|}{2} \right\rceil, \quad |\text{evens}(w)| = \left\lfloor \frac{|w|}{2} \right\rfloor$$

Since both $|\text{odds}(w)| < n$ and $|\text{evens}(w)| < n$ by IH we have

$$|\text{bitrev}(\text{odds}(w))| = |\text{odds}(w)|, \qquad |\text{bitrev}(\text{evens}(w))| = |\text{evens}(w)|$$

That is,

$$\begin{aligned}
|\text{bitrev}(w)| &= |\text{bitrev}(\text{odds}(w)) \cdot \text{bitrev}(\text{evens}(w))| \\
&= |\text{odds}(w)| + |\text{evens}(w)| \\
&= \left\lceil \frac{|w|}{2} \right\rceil + \left\lfloor \frac{|w|}{2} \right\rfloor
\end{aligned}$$

Then we have two cases:

- $|w|$ is even. Then,
$$\left\lceil \frac{|w|}{2} \right\rceil + \left\lfloor \frac{|w|}{2} \right\rfloor = \frac{|w|}{2} + \frac{|w|}{2} = |w|$$
- $|w|$ is odd. Then,
$$\left\lceil \frac{|w|}{2} \right\rceil + \left\lfloor \frac{|w|}{2} \right\rfloor = \left(\frac{|w|+1}{2}\right) + \left(\frac{|w|-1}{2}\right) = |w|$$

Hence, $|\mathrm{bitrev}(w)| = |w|$, and the hypothesis holds!